

THE ECONOMIC VALUE OF USING REALIZED VOLATILITY IN THE INDEX OPTIONS MARKET

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Abstract

We examine the economic benefits of using high frequency volatility measures for pricing, trading and hedging in the S&P 500 index options market. Using the encompassing regression framework, we generate volatility forecasts combining information from long memory high-frequency volatility specifications and option-based implied volatilities. We conduct out-of-sample tests of the volatility forecasts by examining option pricing performance, trading performance based on volatility timing strategies, and the performance of covered options positions for index option writers. Our results support combining forecasts of implied volatility and realized volatility and illustrate that the realized volatility approach has economic value in the context of option pricing and risk management.

Key Words: realized volatility, ARFIMA models, implied volatility, encompassing regressions, combination forecasts, volatility timing, option trading strategies

JEL Classification: G10, G14

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1. Introduction

The extensive research on volatility modeling indicates its importance to asset pricing, portfolio selection, option valuation and risk management. The automation of exchanges and trading processes since the 1990's has led to the widespread availability of high frequency trading data of financial assets and has triggered research on volatility modeling based on intra-day returns. A volatility measure that is gaining increasing acceptance is realized volatility, which is constructed by summing the squares of intra-day returns over very short intervals. Realized volatility is model-free, theoretically free from measurement error and has better accuracy over existing methods in forecasting future volatility (see, Andersen, Bollerslev, Diebold and Ebens, 2001 (ABDE hereafter); Andersen, Bollerslev, Diebold and Labys, 2001 and 2003 (ABDL, 2001 and 2003 hereafter) and Barndorff-Nielsen and Shephard, 2002).

According to the theory of quadratic variation, realized volatility is an unbiased and consistent estimator of return volatility. Under weak regularity conditions, realized volatility will approach true volatility as the returns are sampled at more frequent intervals.¹ Previous research has shown that realized volatility is log normally distributed and daily returns standardized by realized volatility are approximately normal. In addition, realized volatility is found to have long-memory properties and is described by a fractionally integrated autoregressive process. These findings are consistent across

¹ This approach is based on Merton's (1980) earlier work that showed that precise estimates of volatility could be obtained over fixed time intervals as the sampling interval approaches zero. In the limit, unobservable volatility becomes observable.

asset classes (for exchange rates, see ABDEL, 2001 and 2003, for individual stocks see ABDEL, 2001, and for the Dow Jones Industrial Average, see Ebens, 1999).

While these studies have shown that the realized volatility approach is statistically superior in forecasting future volatility, a few others have explored whether this approach provides any economic benefit to its users in investment management and risk management decisions. Fleming, Kirby and Ostdiek, 2003 show that the economic value of using realized volatility in asset allocation decisions for equity markets is substantial for both short and long horizons. They find the overall value of volatility timing using realized volatility to be approximately double the magnitude relative to an ex ante efficient static portfolio. Giot and Laurent (2004) find that an adequately specified ARCH type model (incorporating both kurtosis and skewness) can deliver accurate VaR forecasts and perform as well as a competing VaR model based on the realized volatility. In this paper, we contribute to this line of literature by examining the incremental *economic value* of using realized volatility in the context of option markets. In particular, we explore the economic benefits of using high frequency based realized volatility measures for pricing, trading and hedging in the index options market.

Options traders routinely formulate option volatility forecasts combining information from historical low frequency (i.e. daily, weekly or monthly) return volatilities with implied volatilities obtained from option data (e.g. Canina and Figlewski, 1993). While historical daily volatilities may have generalized autoregressive conditional heteroskedastic (GARCH) type stochastic features, implied volatilities extracted from cross-sectional option data have both moneyness and maturity dimensions. The availability of high frequency data may provide superior volatility

forecasts for option traders by combining intra-day historical volatility and robust implied volatility measures. Previous research based on encompassing regressions has found that a combination of realized volatility and option implied volatility dominates stand-alone implied volatility statistically as a measure of future volatility (Blair, Poon and Taylor, 2001; Li, 2002; Pong, Shackleton, Taylor and Xu, 2004). More recently, Christensen and Nielsen (2006) employ encompassing regression tests to study the information content of option implied volatility relative to both the continuous and jump components of realized volatility. However no prior work has examined the incremental *economic value* of using realized volatility in combination with option implied volatility in the context of option markets.

In addition, difficulties in implementing stochastic volatility option pricing models such as Heston (1993), underscore the importance of combining information from historical data with cross sectional option data. Heston & Nandi (2000) argue that their GARCH based option pricing model has the advantages of combining time series data of spot returns that accommodates volatility clustering features with cross sectional option data, which has forward looking information content. As a result, the GARCH valuation model has fewer parameters to estimate from the option data compared to stochastic volatility pricing models, which leads to a parsimonious estimation procedure. This paper adopts a natural extension of this approach by combining the high frequency volatility measures with the forward-looking option implied volatilities.

In this paper, we examine the economic benefits of using intra-day volatility measures based on the realized volatility approach in the S&P 500 index options market during the period 1996-2002. We focus on at-the-money (ATM) and out-of-the money

(OTM) options with short-term maturities (i.e. 6 to 60 days), as they are most widely traded. Our research design is motivated by two criteria, the use of combination volatility forecasts and the evaluation of the forecasts using out-of-sample tests, and is summarized as follows:

- I. First, we generate volatility forecasts using information from two sources: a) long memory high-frequency volatility specifications involving realized volatility and b) option based implied volatilities in the index options market. We consider four specifications of stand-alone realized volatility in addition to the implied volatility and Glosten, Jagannathan and Runkle (1993) type daily GARCH-leverage volatility (model details are provided in Appendix 1). The simplest specification using high frequency stock index returns is the realized volatility (RV) model, which is the 20-day moving average of realized volatility. We also consider the long-memory autoregressive fractionally integrated moving average (ARFIMA) specification of realized volatility. Following Andersen, Bollerslev and Diebold (2003), we include ARFIMA models that incorporate leverage (ARFIMA-L) and both leverage and jumps (ARFIMA-LJ). The implied volatility (IV) is obtained using the deterministic volatility approach, where a volatility surface is defined as a polynomial function of maturity and moneyness (see Christoffersen and Jacobs, 2004; details in Appendix 2).

In addition to the six stand-alone volatility models, we examine five combination forecasts, arising from combining each historical volatility specification with the option IV.

- II. We next present a method for optimally combining the information from both sources, based on the encompassing regression framework. We suggest that in

exploring the economic significance of RV for option markets, we should be concerned about forecasting IV rather than RV while employing the encompassing regressions; since there is a one-to one-mapping between IVs and option prices for a given maturity, forecasting IVs translates to forecasting prices in the option market.

III. We then investigate the incremental value of using the RVs in practical risk management, based on both static and dynamic performance measures:

- a. We evaluate the static performance of competing volatility forecasting models for options markets for the period 2000–2002 by conducting a comprehensive out-of-sample analysis. We also investigate if combination models lead to better out of sample forecasts of option prices compared to stand-alone models.
- b. We examine whether option investors can trade profitably using delta neutral straddles based on out-of-sample volatility timing trading strategies. We use the framework proposed by Noh, Engle, and Kane (1994) to estimate the profits from daily options trading strategies.
- c. Finally, following Green and Figlewski (1999), we investigate if index option writers such as financial institutions can reduce their risk exposure by pricing and delta hedging daily based on alternate volatility specifications.²

Our empirical evidence on the encompassing regressions shows that volatility forecasting for option pricing should combine information in IV and RV. Combination forecasts involving IV and the simplest RV model even dominate combinations of IV and

daily GARCH models. Combinations of IV and ARFIMA models of RV seem to have the highest explanatory power. Our out-of-sample tests on option pricing performance indicate that a combination of IV and RV measures dominates stand-alone models especially for calls, consistent with the results in the encompassing regressions. In particular, combination of IV and ARFIMA models seem to have the lowest pricing errors.

In addition, our dynamic straddle trading strategies show that a combination of IV and ARFIMA-leverage and jump model of RV has the best performance. In particular, the IV plus ARFIMA-LJ combination has significant average daily returns of 3.23% and a daily Sharpe ratio of 0.15, which is significantly different from the market's Sharpe ratio of -0.0489 . The results are robust to trading costs, bid-ask filters, forecast horizons, and possible non-normalities in underlying returns and suggest that forecasts using IV and ARFIMA models dominate those from stand alone models. Finally, our results indicate that combination volatility forecasts from IV and ARFIMA models of RV help minimize the risk exposure for financial institutions that write options and delta hedge them; the underlying risk though minimized is not eliminated by hedging.

Overall, we find that the RV approach has economic value in trading in the options market and in the context of risk management. Our results strongly support the use of combined forecasts of IV and ARFIMA models of RV in out-of-sample pricing, and trading of index options and suggest that risk exposure from covered options positions can be minimized by using the RV modeling approach. Our findings emphasize the need for future research on closed form pricing of derivatives written on realized

² For the sake of brevity, we present the results only for call options in the subsequent sections, except in section 6, which involves straddle trades. Similarly, we present the tabulated results only at the daily frequency but provide

quadratic variation measures (see for example, Carr et al, (2005) for work in this direction).

The remainder of the paper is organized as follows. In Section 2, we discuss the research design employed in this study. Section 3 describes the data. In section 4, we discuss the results from the encompassing regressions and present the results of our sensitivity tests. Section 5 reports the results from out-of-sample pricing performance. Section 6 presents the results from the volatility based option trading strategies. Section 7 presents the results from sell-hedge positions for option writers and Section 8 provides our summary and conclusions.

2. Related literature and empirical framework

2.1 Generating combination forecasts

Combination forecasts i.e. mixtures of two forecasts have been known to often outperform univariate forecasts (see, Clemen, 1989 for a survey). Diebold and Lopez, 1996 argue that when each candidate model is potentially misspecified, and pooling of underlying information sets is not feasible, individual model forecasts can be combined to generate more informative composite forecasts. Research on volatility forecasting has examined several ways of combining forecasts based on historical volatility and option IV (see Blair, Poon, and Taylor, 2001 and Koopman, Jungbacker, and Hol, 2005). On one hand, historical volatilities are found to have clustering effects in daily or weekly data and long-memory properties (estimated as fractionally integrated processes) in the intra-day data (see, ABDE, 2001). On the other hand, IV is contingent on the

brief discussions of our findings at the weekly and monthly frequencies.

information from the whole IV surface, which is a function of both maturity and moneyness.

2.1.1 Combining IV and GARCH (or stochastic volatility) models

Previous research has examined the information content of IV in the time varying volatility models. For example, Day and Lewis (1992) and Lamoureux and Lastrapes (1993) find that the IV of ATM options has significant information content when incorporated into the GARCH volatility specification for individual stocks. Blair, Poon and Taylor (2001) generate combination forecasts by conditioning the GARCH volatility equation with lagged RV and ATM IV variables. They find that the intra-day equity volatility has marginal incremental information content over the market based IV index (VIX) for one day-ahead forecasts, while VIX dominates at longer time horizons. Koopman et al. (2005) investigate the comparative performance of four different volatility specifications: ARFIMA model for RV, unobserved ARMA components (UC) model for RV, daily stochastic return volatility model augmented with RV and IV, and daily GARCH model augmented with RV and IV. They find that ARFIMA and UC models perform the best for S&P 100 index options (OEX) data. Engle and Gallo (2004) estimate a joint trivariate system of volatilities by combining three variables: demeaned raw squared daily returns, high-low Parkinson volatility and intra-day volatility (RV) and find that RV modeled as a function of lagged variables and corresponding leverage effects produces good forecasts for the VIX. Finally, Donaldson and Kamstra (2005) use combinations of IV and GARCH forecasts along with a trading volume indicator to

predict future volatility and show that in periods of high trading volume, IV has higher information content than GARCH for forecasting future volatility.

2.2 Encompassing regressions with future RV as the dependent variable

An alternative way of combining option IV and historical volatility is by employing an encompassing regression, the traditional workhorse of the volatility forecasting literature. For short-maturity ATM options, the Hull and White (1987) model indicates that IV should represent an unbiased forecast of average volatility over the life of the option and its forecast error should be orthogonal to information at time t . This interpretation forms the basis for the following commonly used log specification of the encompassing regression:

$$\ln(\sigma_{t,T}^{realized}) = \alpha + \beta_1 \ln(\sigma_{t,T}^{FIV}) + \beta_2 \ln(\sigma_{t,T}^{FHV}) + e_{t,T} \quad (1)$$

The variable $\sigma_{t,T}^{realized}$ is obtained by averaging the annualized realized volatilities over the maturity of the option i.e. from day t to day T . The variable $\sigma_{t,T}^{FIV}$ is the forecast of IV on day t using the parameter vector for the IV function estimated on day $t-1$ and the variable $\sigma_{t,T}^{FHV}$ is the forecast of historical volatility conditional on day $t-1$ information, obtained by averaging the daily (annualized) volatility forecasts from a given historical volatility model over the life of the option. Note that both independent variables are conditional on information up to day $t-1$. Tests underling encompassing regressions are subject to the joint null of market efficiency and model adequacy.

Encompassing regressions have been widely used to examine the incremental value of daily and intra-day historical return volatility measures over the option based IV

in forecasting future realized volatilities (for example, Canina and Figlewski, 1993). Encompassing regression tests have to address several methodological issues related to measurement errors in instrumental variables, orthogonal errors in forecasting regressions and biases in regression tests.³ Prior work has attempted to address some of these issues and has shown that IV has significant information content for future volatilities in the index options market. In a recent study, Jiang and Tian (2005) show that model free IV is an efficient forecast for future RV and subsumes all information contained in Black and Scholes IV and past RV in the S&P 500 index options market.

Research based on encompassing regressions suggests that combining intra-day volatility with option IV has better explanatory power for future volatility. Li (2002) using long memory models of RV in foreign currency markets finds that historical data dominates OTC option IVs in volatility prediction. Using foreign exchange data, Pong et al. (2004) show that RV based forecasts dominate GARCH or IV based forecasts for short horizons up to a week, while for longer horizons the IVs perform better. Christensen and Nielsen (2006) find that IV in the S&P 500 index option market has incremental information relative to both the continuous and jump components of RV, subsuming the information content in both of them. Finally, using FX data sampled at different frequencies, Ghysels, Santa-Clara, and Valkanov (2005), find that at weekly and monthly horizons, daily ranges and intra-daily absolute returns dominate RV constructed from intra-day squared returns.

2.2.1 Encompassing regressions with future IV as the dependent variable

³ See, for example, Fleming (1998), Christensen and Prabhala (1998), Poteshman (2000), Neely (2004) and Christoffersen and Mazzotta (2005) for a discussion of these issues.

The encompassing regressions discussed so far have been specified with RV as the dependent variable. As Christensen and Prabhala (1998, p.148) note, “Virtually all studies in the literature have focused on predicting future RV using options data, with less emphasis on the issue of predicting IV .” Since our study focuses on option markets we are mainly interested in forecasting option prices and the corresponding implied volatilities.

We therefore express the encompassing regression with IV as the dependent variable as

$$\ln(\sigma_{t,T}^{IV}) = \alpha + \beta_1 \ln(\sigma_{t,T}^{FIV}) + \beta_2 \ln(\sigma_{t,T}^{FHV}) + e_{t,T} \quad (2)$$

where we target the IV for a $T-t$ maturity option, conditional on the observed market option price at time t and the independent variables are defined as in equation (1). The log volatility specification imposes a non-negative constraint and is likely to be robust to outliers. Specification (2) is similar to a stochastic volatility specification with the IV being part of the latent volatility process. Christensen and Prabhala (1998) suggest that this specification provides a constructive means of forecasting future implied volatilities, and hence future option prices, using variables in the market’s information set. The authors find that such a specification explains an economically significant portion (62%) of the variation in implied volatility. The specification in equation 2 has also been used earlier in the literature to make errors-in variable adjustments in the encompassing regressions (see, Christensen and Prabhala 1998, Christensen and Nielsen, 2006).

By focusing on future IV in the encompassing regressions, we can investigate meaningfully the economic relevance of intra-day RV measure for option markets. Since there is a one-to-one mapping between IV and option prices, by targeting IV at time $t+I$,

we can generate better option price forecasts for option trading.

2.2.2 Estimation and unbiasedness tests for Encompassing Regressions with future IV as the dependent variable

We use the encompassing regression specification in (2) to analyze the unbiasedness and information content of the volatility forecasts. We adopt a research design that minimizes potential measurement errors. The SPX options used in this study are European, have no early exercise possibilities or wild card options (hence justifying the use of the Black-Scholes dividend adjusted pricing model). The non-synchronous measurement problems are attenuated by employing the intra-day option and spot data and using the earliest occurring spot transaction prior to a given option trade as its matching spot data (details provided in Section 3). Obtaining the IV by minimizing market and model prices, through the Implied Volatility Function approach helps address the smile biases and measurement errors (details provided in Section 2.3). Non-overlapping sample estimation has been used in previous studies (e.g. Christensen and Prabhala, 1998) to implement encompassing regressions in order to correct for biases in regression coefficients and their p -values. Implementing non-overlapping sample estimation, however, requires long time series of historical option and spot data (for example, 11.5 years of monthly data in Christensen and Prabhala, 1998, and 16.6 years of monthly data in Christensen and Nielsen, 2006) to ensure adequate size of the sample. Since our main objective in this paper is to forecast implied volatilities from the perspective of the option market participants, we would expect that the recent history of option market information, i.e. a shorter time series, to be more informative. We therefore estimate the encompassing regressions using the Hansen (1982) GMM approach and

employ a heteroscedasticity consistent variance-covariance matrix to minimize the potential overlapping data problems (see Fleming, 1998; Neely, 2004; and Li, 2002). Further, we test and find that our encompassing regressions have no spurious regression issues.

Next we describe the unbiasedness tests. If the information content in the volatility forecasting model is subsumed in the IV forecast, the testable null hypothesis is $\beta_2 = 0$. If the IV forecast is an unbiased estimate of future implied volatility, the testable null hypothesis is $\beta_1 = 1$ and $\beta_2 = 0$. We report Hansen's (1982) t statistics for individual coefficient and the F test statistics for the joint null hypothesis: $\alpha = 0, \beta_1 = 1, \beta_2 = 0$, where the corresponding p -values are robust to overlapping observations. In addition to the adjusted R^2 , we also report the non-nested Diebold-Mariano test statistic and corresponding p -values for each encompassing regression evaluated against the IV benchmark.

2.3 Obtaining historical and IV forecasts

In this section, we describe the historical (daily and intra-day) specifications of volatility and IV measure used in our study. Previous research has found the RV to be a highly persistent process with very slowly decaying autocorrelations (ABDL, 2001 and 2003; ABDE, 2001). Realized volatility does not have unit roots but appears to be fractionally integrated and hence is very slowly mean reverting. Based on the theory of quadratic variation, realized volatility approaches true volatility as the sampling frequency tends to infinity. However, in the presence of jump news events, as the sampling frequency increases, RV tends to the true volatility, which now has both

continuous and jump components (see, Andersen, Bollerslev, and Diebold, 2003; Barndorff-Nielsen and Shephard, 2002). Andersen, et al. (2003) show that the jump component can be isolated from the intra-day data using a non-parametric method, which involves defining a new bi-powered variation measure (BV) and obtaining the jump component by imposing a non-negative constraint on the difference between RV and BV (see Appendix 1 for details).

Based on the stylized facts, we consider the daily GARCH model (as a benchmark specification) and four alternative RV specifications. The five specifications considered for historical volatility are 1) a daily GARCH (1,1) model with leverage effect following Glosten, Jagannathan and Runkle, 1993 (GARCH-L), 2) a 20-day moving average of RV, 3) a long-memory Autoregressive Fractionally Integrated Moving Average (ARFIMA) model for daily RV, 4) an ARFIMA-leverage (ARFIMA-L) model, and 5) an ARFIMA leverage model with jumps (ARFIMA-LJ). The details of model specifications and estimation procedures are provided in Appendix 1. Since the RV is highly persistent, and the logarithmic RV series is observed to be homoskedastic and approximately unconditionally Gaussian, we use the ARFIMA $(I, d, 0)$ representation of the logarithmic RV. We use the autoregressive moving average (ARMA) filter to reduce the impact of spurious serial correlations arising from microstructure frictions (Andersen, Bollerslev, and Diebold, 2003). The leverage component accounts for the asymmetric response of volatility to positive versus negative news shocks. The jump component captures the effects of sudden market news events; incorporating the jump component lowers the impact of lagged RV on days in which the jump component is active thereby lowering the persistence of the RV process.

We estimate each model based on a four-year rolling window for the sample period 1996-2002 and generate multi-step out-of-sample forecasts on a given day for each option according to the underlying maturity horizon.⁴

The IV is obtained using the Implied Volatility Function Approach, based on the method suggested by Christoffersen and Jacobs (2004) and Christoffersen (2003, pp. 138). The IV is obtained using a volatility surface, which is defined in log form as a polynomial function of maturity and moneyness and whose parameters are obtained by minimizing the difference between the observed and Black-Scholes model prices, observed during the 2 to 3 pm time window every trading day. The details are presented in Appendix 2. Also referred to as the ad hoc or practitioner Black-Scholes model, Dumas, Fleming and Whaley (1998) show that this approach outperforms many of the deterministic volatility models. Christofferson (2003) shows that the IV function can be used as a robust benchmark for forecasting purposes.

2.4 Forecasting IV based on a combination of lagged IV and historical volatility

Combination forecasts have a long tradition in the forecasting literature (see Clemen, 1989). Granger and Ramanathan (1984) propose an OLS method for combining multiple forecasts that gives the smallest mean squared error and has an unbiased combined forecast even if individual forecasts are biased.

In addition to the volatility forecasts from each candidate model described in Section 2.3, we use the combination forecasts from each historical volatility specification

⁴ We also examined five-day ahead and twenty-day ahead volatility forecasts corresponding to weekly, and monthly forecasts. For weekly forecasts, we forecast and trade every Wednesday in the out-of- sample strategies, and for monthly forecasts we trade every fifth Wednesday. Only daily forecasts are tabulated, however, due to space constraints.

and implied volatility. The combination forecasts use the information in the IV forecast and the potential incremental information in the historical volatility forecasts. In all therefore we consider eleven volatility measures i.e.: IV, GARCH-L, RV, ARFIMA, ARFIMA-L, ARFIMA-LJ, IV+GARCH-L, IV+RV, IV+ARFIMA, IV+ARFIMA-L and IV+ARFIMA-LJ; the first six refer to stand alone measures and the subsequent ones represent combination forecasts.⁵ We adopt the OLS combination method of Granger and Ramanathan (1984) and follow the forecasting procedure described below ⁶:

- Consider an option (from a specific moneyness category) with a remaining maturity of $T-t$ days, on a given day t . Suppose that we are trying to forecast at time t , the IV for the underlying maturity i.e. $T-t$, denoted as $\hat{\sigma}_{t,T}^{IV}$.
- Using a given time series model, we first obtain the average volatility forecast on day t for the $T-t$ ahead period, conditional on the information set on day $t-k$, where $k=1, 5$ or 20 days, corresponding to daily, weekly or monthly forecasting horizon. We denote the historical volatility forecast as $\sigma_{t,T|\Omega_{t-k}}^{FHV}$.
- Next we obtain the IV forecast on day t for the $T-t$ day ahead period, using the IV function for period $t-k$, where $k=1, 5$ or 20 days, for a given moneyness/maturity option group (as described in Appendix 2). We denote this IV forecast as $\sigma_{t,T|\Omega_{t-k}}^{FIV}$.
- Finally, we make a combined IV forecast at time t , based on the historical and IV forecasts using the forecasting equation

$$\left(\hat{\sigma}_{t,T}^{IV}\right) = \hat{\delta}_0 + \hat{\delta}_1 \left(\sigma_{t,T|\Omega_{t-k}}^{FIV}\right) + \hat{\delta}_2 \left(\sigma_{t,T|\Omega_{t-k}}^{FHV}\right) \quad (3)$$

⁵ Note that the “+” sign refers to the combination of two models.

where the dependent variable $\hat{\sigma}_{t,T}^{IV}$ represents the day t combined volatility forecast for the option (whose maturity is $T-t$ days), conditional on information at day $t-k$. The delta coefficients are calibrated from the previous option and spot market volatility data. Specifically the parameter vector $\hat{\delta} : \hat{\delta} = (\hat{\delta}_0, \hat{\delta}_1, \hat{\delta}_2)$ is obtained by estimating the following regression

$$\left(\sigma_{t,T}^{IV}\right) = \delta_0 + \delta_1 \left(\sigma_{t,T|\Omega_{t-k}}^{FIV}\right) + \delta_2 \left(\sigma_{t,T|\Omega_{t-k}}^{FHV}\right) + e_{t,T} \quad (4)$$

over the previous 20-day window of volatility data, where IV is estimated for a given specific moneyness/maturity group.⁷ For each day t in the sample, forecasts of IV are made using a rolling 20-day window estimation.

2.5 Evaluation of out-of-sample pricing errors

Next, we examine the out-of sample pricing errors of daily option prices computed from volatility forecasts using IV and historical volatility. We use the Black-Scholes model to generate the options prices. The Black-Scholes model is merely used as a tool to provide one-to one mapping between option prices and implicit volatility forecasts and by no means implies that it is the true underlying model (see Jiang and Tian, 2005; Christensen and Nielsen, 2006 for similar arguments). Following previous research (for example, Dumas, Fleming and Whaley, 1998), we employ the following four metrics to compare the pricing performance of alternative volatility models: MSE, the mean of squared price differences; MAE, the mean of absolute price differences;

⁶ We note that the realized volatility model is formulated in terms of the logarithmic volatility. Converting the forecast of log volatility into a pure volatility forecast, as is proposed here, would require a correction term. Since the correction is negligible for short horizons (Andersen et al. 2005), we proceed with the proposed approach.

⁷ Our results are robust when we use longer estimation windows for the volatility data.

ALE, the mean of absolute log price differences; FREQ, the ratio of the total number of trading days on which a particular model has a lower daily MSE than the benchmark IV model.

Christoffersen and Jacobs (2004) point out that the loss function used in parameter estimation and model valuation should be identical when comparing models; otherwise the inferences made will be biased. Since our IV function is estimated using a MSE loss function (see Appendix 2), we would expect the out-of-sample MSE loss function for options prices to be the most reliable metric.

In addition, we examine the correlations among the out-of sample volatility forecasts for candidate models. We consider both calls and puts for alternate forecasting horizons.

2.6 Trading strategies based on competing volatility forecasts

We further investigate if RV forecasts can be used to formulate profitable out-of-sample trading strategies in the options market. We use competing out-of-sample volatility forecasts to trade in short term (i.e. maturity 6-60 days) at-the-money (ATM) delta-neutral straddles. We provide the details of the straddle trading strategy in Appendix 3. We form delta neutral ATM straddles each day, price them based on alternate volatility forecasts and then buy (or sell) them on each day depending on whether they are under (or over) priced. When the straddle is purchased (sold), we assume that we can borrow (invest) funds at a risk-free rate. We apply the bid-ask spread and transaction cost filters for all the trades. We trade only when the absolute price difference between the model and market price is expected to exceed \$0.50; this serves as

a proxy for bid-ask spread costs. We additionally apply 0.5% trading cost, as an estimate of actual trading costs in the market place. The 0.5% trading cost corresponds to a trader with \$48,000 investment who pays a commission amounting to \$120 plus 0.25% of the dollar amount based on a standard commission schedule (see Hull, 2000) p. 160).

As the trading strategies are delta neutral, the returns do not need risk adjustment for minor stock price movements and represent abnormal returns. We compare the profits net of transaction costs and bid-ask spreads. We report the results for the delta neutral straddle trading strategy at daily horizons based on the corresponding stand-alone and combination volatility forecasts. We report the summary statistics for option trading returns along with the robust Sharpe ratios (and the corresponding p -values) taking into account the underlying non-normal distributions following Lo (2002).⁸

2.7 Risk exposure for option writers

Finally, we examine if RV can be used to decrease the risk exposure for option writers based on the competing volatility models. As in Green and Figlewski (1999), we assume that options are written primarily by financial institutions (FIs) and constitute risky assets in their overall diversified portfolio.

We assume that the FI prices an option using a Black-Scholes model based on a given competing volatility forecast each day; the interest rate applicable for pricing the option is obtained by interpolating the two nearest maturities as described in Section 3. Under a covered option position, the FI delta hedges its short options positions using the S&P 500 index at a daily trading horizon. The delta hedge ratios are based on Black-

Scholes Model, conditional on volatility inputs from the candidate volatility. For each moneyness subgroup, the FI writes index options each day and delta hedges the position over the maturity of the option on a daily basis.⁹

As in Bollen and Whaley (2004), the abnormal returns to delta hedging for calls over the life of the option can be expressed as:

$$\frac{\Delta_0 \left(S_T + \sum_{t=0}^T D_t e^{r(T-t)} - S_0 e^{rT} \right) - \left(\text{Max}(0, S_T - K) - c_0 e^{rT} \right) + \sum_{t=0}^{T-1} \Delta_t (S_{t+1} + D_t - S_t) e^{r(T-t)}}{\Delta_0 S_0 - c_0} \quad (5)$$

where Δ_t is the delta value on day t , S_T is the closing value of the index on day T , D_t is the dividend at time t , c_t is the call option value on day t , K is the strike price, r is the risk free rate, subscripts 0 and T refer to the opening and closing days.

Similarly the abnormal returns to delta hedging for puts over the life of the option can be expressed as:

$$\frac{\Delta_0 \left(S_T + \sum_{t=0}^T D_t e^{r(T-t)} - S_0 e^{rT} \right) - \left(\text{Max}(0, K - S_T) - p_0 e^{rT} \right) + \sum_{t=0}^{T-1} \Delta_t (S_{t+1} + D_t - S_t) e^{r(T-t)}}{\Delta_0 S_0 - p_0} \quad (6)$$

where p_t is the put option value on day t . The first term in both the expressions refers to the income from holding the underlying asset until maturity, net of its financing costs. The second term is the gain or loss in the financial position and the third term represents cash flows from marked to market adjustments on a daily basis. The denominator is the cost of the option position at inception. The above abnormal returns for calls and puts are

⁸ Lo (2002) shows that the standard error of the Sharpe ratio for a sample size T is asymptotically given as $\sqrt{\hat{V}_{GMM}/T}$, where $\hat{V}_{GMM}$ is the variance of Sharpe ratio that can be estimated using GMM.

⁹ We also analyze covered strategies for option writers at weekly and monthly trading horizons; this implies that the FI writes options at weekly (every Wednesday) or monthly (5th Wednesday) horizons and delta hedges its position on a weekly or a monthly basis. Delta hedging at weekly or monthly horizons is done using the corresponding horizon volatility forecasts as described in Section 2.3. For weekly (monthly) horizons, we exclude options with maturity less than 5 (20) trading days. We discuss briefly the related untabulated results in Section 7.

adjusted for transaction costs of 0.5% as in straddle trading. We report the summary statistics and the robust Sharpe ratios following Lo (2002) for the percentage daily returns, for the sell-and-hedge strategies for different moneyness subgroups.

3. Data and Summary Statistics

Following the previous literature (for example, Bakshi, Cao and Chen, 1997; Christoffersen and Jacobs, 2004), we choose the S&P 500 index (SPX) options in this study. SPX options are widely traded being the second most liquid options in terms of trading volume and most liquid in terms of open interest in the U.S; further SPX options are easier to value as they are European in nature and have no wild card options.

The SPX intra-day option data for the sample period, January 1996 to December 2002 is obtained from the CBOE. We assume that the S&P 500 daily dividend yield interpolated to match the maturity of the option contract is a reasonable proxy for the dividends paid on each option contract. We choose the final trading hour window for options each day, as it tends to have the most active trading (for example, Jiang and Tian, 2005). For each option trading between 2 pm to 3 pm every day, we identify the corresponding underlying spot index data. We use the earliest occurring spot transaction prior to a given option trade as its matching spot value. Eurodollar interest rates with maturities of one month to 12 months are obtained from Datastream and interest rates for a given option maturity option are obtained by interpolating two Eurodollar rates with the closest maturities (time to maturity is annualized using 365 days convention). We apply the following filters separately to the call and put option data: a). options that violate the arbitrage bounds are dropped, b). options below the minimum tick size of 3/8 are

dropped, c). options with maturities below 6 days are eliminated and d). options with daily volumes less than 100 are dropped. Based on these criteria, our initial sample consists of 86,279 calls and 110,274 puts.

We present the summary statistics of the options data in Table 1. Here we report the average price, moneyness, maturity and IV for options in each moneyness-maturity subgroup for the daily 2 pm–3 pm window data for the seven-year period i.e. January 1996 to December 2002. We report summary statistics for sub samples related to short term (6-60 days), medium term (61-180 days) and long term (> 180 days) maturities and corresponding to three moneyness levels. Out-of-the-money or OTM options are defined as those with moneyness X/S is above (below) 1.03 (0.97) for calls (puts); At-the-money or ATM options as X/S lying between 0.97 and 1.03 and in-the-money or ITM options are those with X/S below (above) 0.97 (1.03) for calls (puts).

The moneyness and maturity related biases are evident from Table 1, for example OTM put implied volatilities dominate ITM put implied volatilities and more so for shorter maturities. Most of the option trading seems to be concentrated in short term ATM and OTM calls and puts. Trades for OTM puts far exceed those of ATM puts, indicating that the former have been widely used as crash insurance. In general, we find that trading is thin for ITM options; and for maturities beyond two months. In our subsequent analysis, we exclude ITM options and all options with medium and long maturities as thin trading could mask the informativeness of prices for these trades and exacerbate the bid-ask spreads. Our final sample consists of 60,825 calls and 76,176 puts.

We use both high frequency and daily S&P 500 data for the 7-year period: January 1996 to December 2002 for a total of 1,738 trading days. The intra-day data is obtained from CBOE, while the daily data is extracted from the Datastream database. The intra-day data is used to calculate RV as the sum of squared returns at finely sampled intervals over a given day added to the square of the previous overnight return as in Blair et al. (2001). Choosing a very high sampling frequency may introduce a bias in the variance estimate due to market microstructure effects (bid-ask bounces, price discreteness or non-synchronous trading). Figure 1 shows the volatility signature plot for RV obtained over a range of sampling frequencies. From the graph we observe that the average RV stabilizes at 15-minute interval, when the bias is minimal. Hence, we choose returns at 15-minute sampling intervals to construct the RV measure.

In Figure 2, we find that RV resembles a long memory process as shown by the autocorrelation function of a fractionally integrated process (ARFIMA). As a result we consider alternative ARFIMA specifications in addition to the daily GARCH-L model as possible candidates for historic volatility (details in Appendix 1); the conditional volatility models are then estimated on each day using the moving window of the previous four years of historical daily data. In our subsequent analysis, we therefore use the first four years of historical index data (viz. 1996-1999) for estimating the volatility models and the last three years (i.e. 2000-2002) for performing the out-of-sample analysis.

4. Results from Encompassing Regressions with IV as the Dependent Variable

The results for the unbiasedness tests for S&P500 (OTM and ATM) options are provided in Table 2. We report the tests for call options only for brevity. The encompassing regression is defined in terms of the log specification as follows: $\ln(\sigma_{t,T}^{IV}) = \alpha + \beta_1 \ln(\sigma_{t,T}^{FIV}) + \beta_2 \ln(\sigma_{t,T}^{FHV}) + e_{t,T}$. This specification helps us understand to what extent IV and hence option prices are predictable.

From the first rows of Panels *A* and *B* in Table 2, we observe that IV has significant information content for the next period implied volatility. The coefficient for IV (and its *t* statistic) declines and the coefficient for historical volatility (and its *t* statistic) increases monotonically from RV to ARFIMA-LJ. The adjusted R^2 is higher for the RV and ARFIMA models compared to the IV model. The ARFIMA volatility measure seems to have a first order impact on the adjusted R^2 ; and adding leverage or jump term leads only to a marginal improvement to the adjusted R^2 measure. Based on the coefficient (and *t* statistic) of the historical volatility and the adjusted R^2 measure, ARIMA-LJ forecasts seem to have the highest additional information content.

The Diebold-Mariano test statistics are significant indicating that IV does not subsume the information contained in the volatility forecast measures for all other models. The *F*-statistic is significant for all the models and indicates that the unbiasedness hypothesis is rejected for OTM and ATM options. The results are consistent across both OTM and ATM options, and suggest that lagged IV is a biased predictor of current implied volatility. The results for puts are qualitatively similar, though IV has a larger information content now relative to calls.

Overall the results show that RV measures have incremental value to the Black-Scholes IV in forecasting future IV in the index option markets. This suggests that volatility forecasting for option pricing should make use of the information in both IV and RV. The model adjusted R^2 s, and the coefficient values (along with underlying t statistics) for historical volatility in general indicate that combinations of IV and ARFIMA models have the highest explanatory power. Further, the IV+ARFIMA combinations also dominate the IV+daily GARCH-L forecasts.

Hence, we use combinations of forecasts in subsequent analysis. In untabulated results, we run the encompassing regression using the instrumental variable method to correct for error in variable bias as in Christensen and Nielsen (2006) and find the results remain unchanged.

We conclude, based on results in this section, that the natural approach to forecast IV for options markets is to harness the information from both ARFIMA models of intra-day RV and IV from the cross-sectional option data.

5. Tests of Pricing Performance

In this section, we examine the pricing efficiency of alternative volatility models. In Table 3, we report the summary statistics for daily volatility forecasts for the competing models for the call option data. From Panel A, we observe that in general, the average volatility forecasts from the RV and ARFIMA models are lower than those from the IV and GARCH-L models for both OTM and ATM options. Though the untabulated results for puts are qualitatively similar, the put IV forecasts are much higher compared to the call IV forecasts. This could arise if the prices of puts (especially OTM puts) are

influenced by hedging demand. In Panel *B*, we observe that the volatility forecasts from the IV model have a higher correlation with volatility forecasts from the ARFIMA models than with the GARCH-L model, implying that long memory intra-day volatility measures are more sensitive to market new events compared to low frequency GARCH models.

Table 4, Panel *A* presents the average market prices and Black-Scholes model prices for call options based on one-day ahead forecasts from competing volatility models. For both OTM and ATM options, models prices for the IV model seem to be on average closest to market prices. The RV and ARFIMA models generate the lowest mean and median option prices. However, the combined volatility forecasts (from IV and intra-day models) have means and medians closer to those observed in the market. The volatility of option prices based on IV (stand alone historical volatility models) forecasts are much higher (lower) compared to the volatility of market prices; the combination models however generate prices whose standard deviations are closer to those observed in the market.

In Panel *B*, we present the comparative metrics for pricing errors to evaluate the models. Since the IV function is estimated using a MSE loss function, we expect the out-of-sample MSE loss function for options prices to be the least biased (Christoffersen and Jacobs, 2004). In general, OTM call options have lower pricing errors than ATM calls. The results indicate that prices based on combined forecasts that use the information in both IV and historical volatility models have the lowest pricing errors compared to stand-alone models. For OTM options, the IV+ARFIMA combinations outperform all other models based on the MSE, MAE, ALE and FREQ metrics. Among the IV+ARFIMA

combinations, the IV+ARFIMA-LJ model combination has the best performance based on all the four metrics. This implies that not only does the IV+ARFIMA-LJ combination have the lowest MSE, but also on most days its MSE is also lower than that of the benchmark IV model. For ATM options, the IV and daily GARCH combinations have the lowest MSE and MAE followed by the IV+ARFIMA-LJ combination. The FREQ measure indicates that IV+ARFIMA combinations have the best performance.

In untabulated results, we find that for puts, IV has the most information content for OTM options. The pricing errors for puts based on the MSE, MAE and ALE metrics are higher in magnitude relative to calls, especially for the OTM options. This is not surprising since OTM puts are in high demand as crash insurance and therefore are more expensive; their prices are hard to justify solely based on historical volatilities. For ATM puts, the combination of IV+ARFIMA models outperform all others; the IV+daily GARCH combinations have much higher MSE though they perform equally well in other metrics.

We also find qualitatively similar results for call and put options prices based on weekly and monthly volatility forecasts, with forecast errors monotonically increasing at these intervals. For example for calls, at weekly (monthly) intervals, IV+ARFIMA-LJ (IV+ARFIMA) model tends to have the lowest forecast error metrics. For puts, IV+ARFIMA models perform well closely followed by IV+GARCH-L models for ATM options, while IVs continue to dominate all others for OTM options.

In summary, out-of-sample pricing results at daily, weekly and monthly horizons indicate that combination of IV and ARFIMA models have the lowest pricing errors on average and on most of the days. The results in this section are supportive of

combination forecasts and are consistent with the findings from the encompassing regression that volatility forecasting for option pricing should incorporate the information in both IV and RV measures.

6. Option Trading Strategies

Table 5 presents the results for daily trading strategies using ATM delta-neutral straddles based on volatility forecasts. The procedure is described in Appendix 3. The average moneyness of straddles (X/S) is almost one, while the average maturity is about 20 days. The IV+ARFIMA-LJ model has the closest average model price for straddles compared to the market.

The number of buys and sells of the delta-neutral straddles are reported using a bid-ask filter (\$0.50) for differences between model and market prices. We find that the total trades are different across models because of attrition due to the filter rule. Straddles are bought more often than sold for the IV model and the combined forecasts models since these model prices are higher than the market prices. On the other hand, for the stand-alone RV models, the trades are predominantly short straddles.

Next, we report percentage daily returns on trading in the delta-neutral straddles using a bid-ask filter and 0.5% transaction costs. As the trading strategies are delta neutral, the returns do not need risk adjustment for minor stock price movements and represent abnormal returns. Besides the summary statistics for profits from straddle trading, we present the first four moments of returns and the p -values for the mean return. The p -values for the mean are based on the modified t -test (Johnson, 1978) that accounts for the asymmetric distributional property of the returns from option trading strategies.

This test statistic has been used by Bollen and Whaley (2004) in their analysis of simulated delta-neutral trading strategies. The test-statistic is given by

$$t_j = \left(RET + \frac{\sigma S}{6n} + \frac{RET^2 S}{3\sigma} \right) (\sigma^2 n)^{-0.5} \quad (7)$$

where RET is the mean return, σ is the standard deviation, S is the skewness and n is the number of observations.

We also report the ratio of positive return vs. negative return days. Finally, we present the Sharpe ratio and the corresponding p -values robust to non-iid returns, estimated using a GMM framework following Lo (2002).

The results in Table 5 results show that the IV-ARFIMA combination models have higher mean returns (statistically significant at the 1% level) than stand-alone models, with the IV- ARFIMA-LJ model having the highest return of 3.23%. The returns from the IV-ARFIMA combination models also in general have much higher (positive) skewness and kurtosis compared to the other models. The stand-alone RV models have the lowest average returns. Stand-alone RV models also have a high ratio of positive return vs. negative return days, with stand-alone ARFIMA- LJ model outperforming the rest.

Further in terms of the Sharpe ratio, IV-ARFIMA combination models have higher risk adjusted returns than the stand-alone models as well as the spot index market. The IV+ARFIMA-L and IV+ARFIMA-LJ models have the highest daily Sharpe ratio of 0.15, being significantly different from zero at a 1% level. The corresponding daily Sharpe ratio for the market during the period 2000-2002 was -0.0489 .

In untabulated results, we find that combination models once again dominate the stand-alone models based on mean returns and Sharpe ratios for trading at weekly and monthly frequencies. For weekly returns, the IV+ARFIMA-LJ model has the highest weekly return of 8.67% and a Sharpe ratio of 0.26 that is significantly different from zero at a 1% level. For monthly returns, the IV+RV, and the two IV+ARFIMA models have the highest monthly returns of 10.96% and a Sharpe ratio of 0.41, which is significantly different from zero at a 5% level of significance.¹⁰

In summary, the combination forecasts using IV and RV measures dominate the other models in trading strategies at all trading frequencies. In particular, the out-of-sample trading results indicate that the combination of IV and ARFIMA models of RV have the highest Sharpe ratios. Our results on combination forecasts indicate that there is incremental information in forecasts of ARFIMA models of RV that can lead to higher and statistically significant Sharpe ratios, even after controlling for possible non-normalities in the underlying returns.

7. Option Writing

Finally, we examine the performance of covered options positions of FIs; we assume that FIs write options and dynamically delta hedge their positions, both based on forecasts from competing volatility models. Given the bear market run of 2000-02, we should expect positive (negative) returns for sell-hold call (put) positions. For FIs who hold covered positions, however, we should expect delta hedging to lower (increase) the returns for short call (put) positions, before any transaction costs. In Table 6, we report

¹⁰ Santa-Clara and Saretto (2005) point out that the average round-trip costs of near-maturity ATM calls and puts is

the daily returns net of trading costs from delta hedging daily the short positions for calls. In general, we notice that sell-delta hedge strategies yield negative returns and are not profitable. However the combination forecasts from IV and ARFIMA models help FIs control their losses. For OTM call writers who delta hedge, while all models yield negative and significant returns, the IV+ARFIMA- LJ combination generates the lowest negative daily returns (or lowest losses) of -0.008%, which are not significantly different from zero. Similarly while the Sharpe ratios from all the volatility models are negative and not significant, IV+ARFIMA- LJ combination has the lowest negative Sharpe ratio (or lowest loss exposure) of -0.057.

For ATM call writers, the IV-ARFIMA and IV-ARFIMA-L combinations generate lowest losses of -0.026%, which are significant at the 5% level and the lowest loss exposure in terms of a Sharpe ratio of -0.117, which is not significant at the 5% level.

We find somewhat similar results for OTM (ATM) puts (not reported in the tables). The IV-ARFIMA-LJ (IV-RV) model has a daily return of -0.141% (-0.048%) and the highest Sharpe ratio of -0.329 (-0.217), which is not significantly different from zero at the 5% level. We also notice that Sharpe ratios for covered put positions are more negative relative to those from covered call positions, indicating that the former could have been less attractive to FIs in our sample period. Overall, we find that the IV+ARFIMA models combinations help marginally lower the risk exposure of covered call and put positions based on the Sharpe ratios particularly at daily and weekly horizons.

around 5.5%. With these trading costs, our volatility timing strategies are profitable only at weekly and monthly frequencies.

8. Summary and Conclusions

In this paper, we examine the incremental *economic value* of using realized volatility in combination with option implied volatility for pricing, trading and hedging in the S&P 500 index options market.

We model RV as a long memory process. We compare four specifications of RV to a robust IV measure and daily GARCH-leverage volatility. In addition, we study forecasts that combine the information from RV and IV forecasts; in all we consider eleven competing specifications. We focus on the most liquid segment of the options market viz., ATM and OTM options with 6-60 day maturities. In order to explore the relevance of RV for option markets, we forecast IV employing the encompassing regressions. Our findings based on out-of-sample tests, can be summarized as follows:

- The encompassing regression tests show that RV has incremental value over IV in forecasting future implied volatility. Our out of sample pricing performance results illustrate that combinations of IV and ARFIMA models of RV have the highest explanatory power.

Next, we investigate the incremental value of using the RV based on both static and dynamic performance measures:

- The out-of-sample pricing results indicate that combinations of IV and ARFIMA models of RV have the lowest pricing errors on average and on most days compared to the other models, consistent with the results in the encompassing regressions.
- Trading of delta neutral ATM straddles indicates that a mixed model dominates in volatility timing strategies at the daily level, with significant average daily returns

of 3.23% after adjusting for trading costs. We also find that the combination of IV and ARFIMA models of RV have the highest Sharpe ratios and our results are robust to trading costs, bid-ask filters, forecast horizons, and possible non-normalities in underlying returns.

- Finally, dynamic covered option positions indicate that combination volatility forecasts from IV and ARFIMA models of RV create the least risk exposure for financial institutions that write call options and delta hedge them, but this risk though minimized is not eliminated by hedging. Our findings are robust to transaction costs, bid-ask filters, and forecast horizons.

Overall, the results suggest that intra day volatility measures can be combined with robust IV forecasts to gainfully trade and hedge in the option markets. The RV approach therefore has economic value in trading in the options market and in the context of risk management. Options traders formulate option volatility forecasts combining historical volatilities with option implied volatilities. In addition, research on GARCH based option pricing models underscores the importance of combining information from historical data with cross sectional option data. This paper adopts a natural extension of this approach by combining the high frequency volatility measures with the forward-looking option implied volatilities. Empirical evidence in this paper substantiates the need for future research on closed form pricing of derivatives written on realized quadratic variation measures. This follows the recent developments in the OTC variance option and swap markets, where participants can speculate and hedge in the underlying quadratic variation measures. Future research should also investigate the robustness of using high frequency volatility measures for hedging risks in the futures markets.

Appendix 1: Model Summary

1. GJR-GARCH(1,1) (GARCH-L model)

$$r_t = \mu + \varepsilon_t, \quad h_t = \omega + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta h_{t-1}$$

$$\varepsilon_t = \sqrt{h_t} z_t, \quad z_t \sim N(0,1)$$

- r_t = daily returns on day t
- “ I ” is an indicator function equal to 1 if the argument is true, zero otherwise.

2. RV

Realized volatility is expressed as a 20- day arithmetic average of past RVs

3. ARFIMA model

$$(1-L)^d \ln \sigma_t^2 = \omega + \gamma \ln \sigma_{t-1}^2 + u_t$$

- Daily realized variance(RV) = $\sigma_t^2 = \sum_{j=1}^{1/\Delta} r_{t+j\Delta,\Delta}^2$
- Discrete sampled Δ -period returns $r_{t\Delta} = \ln p_t - \ln p_{t-\Delta}$

4. ARFIMA with Leverage Effect (ARFIMA-L model)

$$(1-L)^d \ln \sigma_t^2 = \omega + \gamma \ln \sigma_{t-1}^2 + \phi \ln \sigma_{t-1}^2 I(r_{t-1} < 0) + u_t$$

5. ARFIMA with Leverage Effects and Jumps (AFIMA-LJ model)

$$(1-L)^d \ln \sigma_t^2 = \omega + \gamma \ln \sigma_{t-1}^2 + \phi \ln \sigma_{t-1}^2 I(r_{t-1} < 0) + \beta_j J_{t-1} + u_t$$

- “ J ” is a jump measure using the high frequency data with the theory of quadratic variations (see Andersen et al. 2003).

$$J_t = \max[RV_t - BV_t, 0]$$

- BV refers to realized bi-power variation

$$\bullet \quad BV_t = \mu_1^{-2} \sum_{j=2}^{1/\Delta} \left| r_{t+j\Delta,\Delta}^2 \right| \left| r_{t+(j-1)\Delta,\Delta}^2 \right| \quad \text{with } \mu_1 = \sqrt{2/\pi}$$

All the ARFIMA models are based on daily realized volatilities. Further, the calculation of realized volatilities includes the overnight adjustment. Specifically, the first return on a given day is calculated as the log difference of the first trade that day and the closing index from the previous trading day. Similarly the second return on given day is obtained as the log difference between the first trade and the one at the 15 minute grid.

Appendix 2: Implied Volatility Function Approach

1. Calculate the implied volatility (IV) using the Black-Scholes model for all observed calls (or puts) on a given day t , after applying the following filters to the option data: a). options outside the upper and lower boundaries are dropped, b). options below the minimum tick size of 3/8 are dropped, c). option below 6 day maturities are eliminated and d). options with implied volatility (IV) higher than 2 are dropped.

2. Define log volatility on day t as a polynomial function of that day's moneyness and maturity as below:

$$\ln \sigma_{t,i}^{IV} = \alpha_0 + \alpha_1 \left(\frac{X_{t,i}}{S_t} \right) + \alpha_2 \left(\frac{X_{t,i}}{S_t} \right)^2 + \alpha_3 (mat_{t,i}) + \alpha_4 (mat_{t,i})^2 + \alpha_5 \left(\frac{X_{t,i}}{S_t} \right) (mat_{t,i}) \quad (1)$$

$\forall i = 1, 2, \dots, N$ on day t

where $\ln \sigma_{t,i}^{IV}$ is the day t log implied volatility for option i , $X_{t,i}$ is strike price for option i on day t , S_t is the underlying spot value on day t , $mat_{t,i}$ is the annualized maturity of the option i on day t and N refers to the total number of call (or put) options available on day t . The log function ensures non-negative volatility values.

3. Obtain the parameter vector α , where $\alpha = \{\alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5\}$, on a given day t for the polynomial (1) by minimizing the MSE of the market and Black-Scholes model prices for all N number of observed calls (or puts) on that day i.e.:

$$\text{Min}_{\alpha} \left\{ \frac{1}{N} \sum_{i=1}^N [mktpr_{t,i} - BSpr_{t,i}(S_t, X_{t,i}, r_t, mat_{t,i}, \sigma_{t,i}^{IV}(S_t/X_{t,i}, mat_{t,i}, \alpha))^2] \right\} \quad (2)$$

where Mkt and BS refer to observed and Black-Scholes model option prices respectively. The model price for option i on a given day t is a function of corresponding strike price, annualized maturity and that day's spot price (S_t), risk free rate (r_t) and latent volatility, which is in turn a function of moneyness, maturity and parameter vector of the polynomial.

4. Next, suppose that we have an option trading on day t with a given moneyness m and maturity of $T-t$ days. We then condition on day $t-k$ parameter vector α (where α is obtained from step 3 using specific moneyness m and maturity $T-t$ as inputs), and forecast IV on day t for the $T-t$ day horizon; k here refers to the forecasting horizon equal to one, five, or twenty days corresponding to daily, weekly or monthly forecasts.
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Appendix 3: Trading strategy based on straddles

1. Consider an option trading with $T-t$ days to maturity on day t . Obtain the volatility forecast for $T-t$ horizon on day t , for each volatility model, as described in Sections 2.3 and 2.4. Use the volatility forecast to price an option on day t using the Black-Scholes pricing formula.
2. From the in-sample daily volatility forecast for day t from step 1, obtain the delta-neutral (DN) straddle for that day. In particular, given that delta of the call and put are respectively $N(d_1)$ and $N(d_1) - 1$, the strike price X that gives a delta -neutral straddle on day t can be solved as :

$$N(d_1) + N(d_1) - 1 = 0; \quad x = \frac{S}{e^{\left(-r_f - \frac{\sigma^2}{2}\right)(T-t)/365}}$$

where S, r_f, σ and $(T-t)/365$ refer to the spot price, risk free rate, volatility and annualized option maturity respectively. On day t , determine that traded straddle combination whose exercise price is closest to the X described above. Use only ATM calls and puts in this exercise. Price the ATM-DN straddle using the Black-Scholes option-pricing model.

3. Next, buy or sell the ATM-DN straddle on day t depending on whether the straddle model price is under or over-priced relative to the closing market straddle price on that day.

The rate of return is calculated as follows:

$$\text{Return on buying a straddle} = \frac{C_t + P_t - C_{t-1} - P_{t-1}}{C_{t-1} + P_{t-1}} - r_f$$

$$\text{Return on selling a straddle} = \frac{-(C_t + P_t - C_{t-1} - P_{t-1})}{C_{t-1} + P_{t-1}} + r_f$$

where C_t and P_t refer to the call and puts prices of S & P 500 index options. When the straddle is purchased, we assume that the agent can borrow at a risk-free rate. Similarly, when the straddle is sold, we let the agent invest the proceeds in a risk-free rate.

4. Implement steps 1-3 for each trading day in the sample.
 5. For all the trades, apply the bid-ask spread filter (that accounts for the bid-ask bounce) as in Noh et al. (1994). The agent trades only when the absolute price difference between the model and market price is expected to exceed \$0.50.
 6. Further, apply 0.5% trading costs, while calculating the net returns for the strategies. The 0.5% trading cost corresponds to a trader with \$48,000 investment who pays a commission amounting to \$120 plus 0.25% of the dollar amount according to a standard commission schedule (see Hull (2000) p. 160).
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Table 1

Sample statistics for the S&P 500 Index option data (Jan 1996- Dec 2002)

For each option that trades between 2:00-3:00 pm every day, we identify the corresponding spot index value based on the most recent spot transaction. We obtain the underlying interest rate by interpolating two Eurodollar rates with the closest maturities (time to maturity is annualized using the 365 days convention). We apply the following filters separately to the call and put option data: a). options that violate the arbitrage bounds are dropped, b). options below the minimum tick size of 3/8 are dropped, c). options below 6 day maturities are eliminated and d). options with implied volatility (IV) higher than 2 are dropped. The IVs are calculated using the IV surface as described in Appendix 2. The IVs are calculated using the IV surface as described in Appendix 2. For each moneyness sub-sample, we report the number of observations and averages for option price, maturity, moneyness and volatility for calls in Panel A and for puts in Panel B.

		Panel A. CALL OPTIONS (Total trades: 86,279)				Panel B. PUT OPTIONS (Total trades: 110,306)				
		Maturity			All Maturities		Maturity			All Maturities
		6-60 days	60-180 days	≥ 180 days			6-60 days	60-180 days	≥ 180 days	
Nobs		29,978	5,678	773	36,429		45,144	11,406	1,509	58,059
avg price		5.11	12.81	26.28	6.27		5.07	11.14	21.39	6.01
avg. maturity	OTM: X/S ≥ 1.03	28.3	94.58	244.04	41	OTM: X/S ≤ 0.97	28.24	99	231.99	43.82
avg. moneyness		1.06	1.1	1.15	1.07		0.9	0.85	0.84	0.89
avg. IV		0.19	0.19	0.23	0.19		0.29	0.27	0.27	0.29
Nobs		31,847	2,290	93	34,230		31,032	3,069	273	34,374
avg price		14.67	35.64	68.85	13.76		14.68	32.98	54.46	14.27
avg. maturity	ATM: 0.97< X/S <1.03	21.44	88.8	241.9	21.04	ATM: 0.97< X/S <1.03	22.11	92.36	244.11	24.86
avg. moneyness		1.01	1.01	1.01	1.01		0.99	0.99	1	0.99
avg. IV		0.19	0.2	0.18	0.2		0.19	0.2	0.29	0.19
Nobs		1,162	231	18	1,411		782	373	95	1,250
avg price		75.34	126.59	120.66	81.2		64.53	101.74	108.7	77.68
avg. maturity	ITM: X/S ≤ 0.97	26.07	91.84	231.56	35.34	ITM: X/S ≥ 1.03	27.85	98.87	250.22	62.6
avg. moneyness		0.94	0.91	0.92	0.94		1.05	1.08	1.07	1.06
avg. IV		0.27	0.31	0.32	0.28		0.21	0.21	0.27	0.22
Nobs		62,987	8,199	884			76,958	14,848	1,877	
avg price		10.96	22.15	32.57			9.09	17.77	30.62	
avg. maturity	All moneyness	24.65	92.86	243.39		All moneyness	25.59	97.47	234.68	
avg. moneyness		1.04	1.07	1.13			0.94	0.89	0.87	
avg. IV		0.19	0.2	0.22			0.25	0.26	0.28	

Table 2

Encompassing regression tests with actual implied volatility as the dependent variable (S&P 500 Index call options: Jan 1996- Dec 2002)

The sample consists of out-of-the-money (OTM) call options defined as $X/S \geq 1.03$ and at-the-money (ATM) call options defined as $0.97 < X/S < 1.03$ with maturity ranging from 6 – 60 days. For each sub-sample, historical volatility (HV) forecasts for five different models are obtained as described in Appendix 1. To obtain implied volatility (IV), we first apply option filters as described in Table 1. Then for each sub sample, we identify the last option trading during the 2 pm-3 pm window each day and use the parameter vector for the IV function estimated on day $t-1$ to forecast IV for next day i.e. day t , where t represents each day in the sample (details of IV estimation are described in Appendix 2). The encompassing regression is defined as: $\ln(\sigma_{t,T}^{IV}) = \alpha + \beta_1 \ln(\sigma_{t,T}^{FIV}) + \beta_2 \ln(\sigma_{t,T}^{FHV}) + e_{t,T}$. Results are reported for the joint null hypothesis: $\alpha = 0, \beta_1 = 1, \beta_2 = 0$. FHV and FIV refer to the volatility forecasts generated respectively from the historical volatility and implied volatility function approaches, conditional in the information set at time $t-1$. The historical volatility forecasts are from the following models and are described in Appendix 1: generalized autoregressive heteroskedasticity model with leverage (GARCH-L), realized volatility (RV), autoregressive fractionally integrated moving average (ARFIMA), ARFIMA model with leverage effect (ARFIMA-L) and the ARFIMA model with leverage and jump (ARFIMA-LJ). The volatility forecasts are for the maturity date, T using information from day t . The dependent variable $(\sigma_{t,T}^{IV})$ represents the actual IV calculated from the option on day t . The independent variable $(\sigma_{t,T}^{FIV})$ is the forecast of IV on day t using the parameter vector for the IV function estimated on day $t-1$. The independent variable $(\sigma_{t,T}^{FHV})$ is the forecast of HV obtained by averaging the daily (annualized) volatility forecasts from a given historical volatility model over the life of the option. The encompassing regressions are estimated using the Hansen's GMM procedure and is robust to overlapping observations. For each regression, we report Hansen's (1982) t statistics for individual coefficients (in parentheses) and the adjusted R^2 . The F -statistic (p -values in parentheses) tests the joint null hypothesis: that $\alpha = 0, \beta_1 = 1, \beta_2 = 0$. We also report the Diebold-Mariano (DM) test statistic for each encompassing regression (with the corresponding p -values in parentheses) against the IV benchmark. We report the results for OTM call option in Panel A and for ATM call options in Panel B.

Panel A: OTM: $X/S \geq 1.03$ ($N = 724$)						
Volatility forecast model	α	β_1	β_2	Adj- R^2	DM	F test
IV	-0.217 (19.099)	0.857 (120.979)		0.778		38.15 (0.0001)
GARCH-L	-0.16 (9.606)	0.805 (77.593)	0.091 (8.536)	0.781	9.25 (0.0001)	29.11 (0.0001)
RV	-0.116 (7.445)	0.786 (80.889)	0.120 (13.895)	0.785	19.63 (0.00)	34.19 (0.00)
ARFIMA	-0.032 (2.358)	0.667 (77.748)	0.282 (35.676)	0.798	13.6 (0.0001)	52.97 (0.0001)
ARFIMA-L	-0.016 (1.24)	0.655 (80.716)	0.302 (40.334)	0.801	13.34 (0.0001)	56.46 (0.0001)
ARFIMA-LJ	-0.013 (1.034)	0.653 (81.49)	0.306 (41.434)	0.801	13.28 (0.0001)	57.15 (0.0001)
Panel B: ATM: $0.97 < X/S < 1.03$ ($N = 713$)						
IV	-0.178 (14.977)	0.879 (110.224)		0.796		26.48 (0.0001)
GARCH-L	-0.116 (6.559)	0.837 (70.814)	0.079 (7.055)	0.798	10.23 (0.0001)	20.72 (0.0001)
RV	-0.044 (2.811)	0.806 (77.425)	0.135 (15.672)	0.803	24.83 (0.00)	28.07 (0.00)
ARFIMA	0.044 (3.272)	0.709 (79.052)	0.273 (35.547)	0.816	22.82 (0.0001)	46.47 (0.0001)
ARFIMA-L	0.057 (4.219)	0.702 (78.180)	0.286 (37.167)	0.817	22.81 (0.0001)	48.59 (0.0001)
ARFIMA-LJ	0.059 (4.409)	0.700 (77.907)	0.289 (37.599)	0.818	22.86 (0.0001)	49.15 (0.0001)

Table 3

Descriptive statistics and correlations of daily volatility forecasts (S & P 500 Index Call options: Jan 1996-Dec 2002).

The sample consists of out-of-the-money (OTM) call options defined as $X/S \geq 1.03$ and at-the-money (ATM) call options defined as $0.97 < X/S < 1.03$ with maturity ranging from 6 – 60 days. For the two sub-samples, we calculate out-of sample daily volatility forecasts for the IV and five other model specifications (described in Appendix 1) and report the corresponding summary statistics for in Panel A. In Panel B, we present correlations for one-day ahead volatility forecasts from different models.

<i>Panel A: Descriptive statistics of volatility forecasts</i>		IV	GARCH -L	RV	ARFIMA	ARFIMA-L	ARFIMA-LJ
OTM	Mean	0.21	0.22	0.17	0.18	0.18	0.18
	Median	0.20	0.20	0.16	0.17	0.17	0.17
	Standard deviation	0.05	0.04	0.04	0.04	0.04	0.04
	Maximum	0.43	0.42	0.34	0.32	0.31	0.31
	Minimum	0.09	0.14	0.08	0.12	0.12	0.12
ATM	Mean	0.24	0.22	0.17	0.18	0.18	0.18
	Median	0.22	0.20	0.16	0.18	0.18	0.18
	Standard deviation	0.06	0.04	0.04	0.04	0.04	0.04
	Maximum	0.47	0.42	0.34	0.32	0.31	0.31
	Minimum	0.11	0.14	0.08	0.12	0.12	0.12
<i>Panel B: Correlations among one-day ahead volatility forecasts</i>		IV	GARCH -L	RV	ARFIMA	ARFIMA-L	ARFIMA-LJ
OTM	IV	1.00	0.74	0.66	0.81	0.81	0.81
	GARCH-L	0.74	1.00	0.66	0.69	0.70	0.70
	RV	0.66	0.66	1.00	0.77	0.77	0.77
	ARFIMA	0.81	0.69	0.77	1.00	0.99	0.99
	ARFIMA-L	0.81	0.70	0.77	0.99	1.00	0.99
	ARFIMA-LJ	0.81	0.70	0.77	0.99	0.99	1.00
ATM	IV	1.00	0.69	0.64	0.77	0.77	0.78
	GARCH-L	0.69	1.00	0.65	0.69	0.70	0.70
	RV	0.64	0.65	1.00	0.78	0.78	0.78
	ARFIMA	0.77	0.69	0.78	1.00	0.99	0.99
	ARFIMA-L	0.77	0.70	0.78	0.99	1.00	0.99
	ARFIMA-LJ	0.78	0.70	0.78	0.99	0.99	1.00

Table 4.

Comparison of alternative volatility models based on daily price forecasts (S & P 500 Index Call options: Jan 1996 – Dec 2002)

The sample consists of out-of-the-money (OTM) call options defined as $X/S \geq 1.03$ and at-the-money (ATM) call options defined as $0.97 < X/S < 1.03$ with maturity ranging from 6 – 60 days. In Panel A, we report summary statistics for one-day ahead call option prices based on volatility forecasts from different models. We also consider combination forecasts where we combine IV with volatilities from alternate models. The combination forecasts are obtained using the encompassing regression specification: $\ln(\sigma_{t+l,T}^{IV}) = \alpha + \beta_1 \ln(\sigma_{t,T}^{FIV}) + \beta_2 \ln(\sigma_{t,T}^{FHV}) + e_{t,T}$, based on the previous 20 trading days of data. In Panel A, we report the summary statistics, and in Panel B, we report out-of sample performance metrics for one-day ahead call option price forecasts. MSE is the mean of squared price differences; MAE is the mean of absolute price differences; ALE is the mean of absolute log price differences and FREQ is ratio of total # of days on which a particular model has a lower daily MSE than the benchmark i.e. the IV model.

Panel A: Summary statistics for one-day ahead call option price forecasts

		Market prices	IV	GARCH- L	RV	ARFIMA	ARFIMA-L	ARFIMA-LJ	IV+GARCH-L	IV+RV	IV+ARFIMA	IV+ARFIMA-L	IV+ARFIMA-LJ
OTM	Mean	5.97	6.07	6.59	3.62	4.06	4.06	4.07	5.99	6.00	5.98	5.96	5.96
	Median	4.30	4.65	5.23	2.19	2.95	3.00	3.00	4.63	4.56	4.71	4.65	4.64
	Standard Deviation	5.12	5.13	5.39	4.01	3.71	3.71	3.71	4.86	4.86	4.85	4.84	4.84
	Maximum	30.50	30.51	44.47	30.32	21.14	21.40	21.48	27.55	25.51	27.13	26.84	26.76
	Minimum	0.40	0.04	0.02	0.01	0.02	0.02	0.02	0.22	0.20	0.21	0.20	0.20
ATM	Mean	23.39	23.43	21.31	16.29	17.40	17.41	17.42	23.35	23.51	23.58	23.56	23.56
	Median	22.00	21.37	19.67	14.21	15.69	15.71	15.78	22.02	22.20	22.25	22.10	22.07
	Standard Deviation	12.65	13.00	11.62	10.47	10.37	10.34	10.34	12.03	12.12	12.23	12.21	12.21
	Maximum	78.00	78.90	91.50	72.98	68.77	68.70	68.53	74.11	73.26	74.24	74.19	74.20
	Minimum	2.50	2.59	1.44	0.90	1.18	1.16	1.17	2.99	3.43	3.23	3.27	3.27

Table 4 continued:

Comparison of alternative volatility models based on daily price forecasts (S & P 500 Index Call options: Jan 1996 – Dec 2002)

Panel B: Pricing Performance for Alternate Volatility Models Based on Daily Forecasts

		IV	GARCH -L	RV	ARFIMA	ARFIMA-L	ARFIMA-LJ	IV+GARCH- L	IV+RV	IV+ ARFIMA	IV+ ARFIMA-L	IV+ ARFIMA-LJ
OTM	MSE	2.693	8.829	13.623	9.004	8.867	8.813	2.521	2.571	2.379	2.383	2.379
	MAE	1.004	2.023	2.706	2.061	2.053	2.048	1.018	1.029	0.985	0.982	0.981
	ALE	0.238	0.439	0.895	0.571	0.566	0.563	0.216	0.217	0.21	0.211	0.211
	FREQ	0	0.277	0.169	0.228	0.217	0.22	0.5	0.492	0.505	0.515	0.517
ATM	MSE	9.65	30.93	75.30	53.93	53.68	53.47	6.79	7.68	7.58	7.51	7.47
	MAE	2.17	3.97	7.27	6.04	6.03	6.02	1.78	1.88	1.86	1.85	1.85
	ALE	0.10	0.20	0.43	0.33	0.33	0.33	0.09	0.09	0.09	0.09	0.09
	FREQ	0.00	0.32	0.12	0.14	0.14	0.14	0.58	0.58	0.59	0.59	0.59

Table 5

Daily trading with at-the-money (ATM) delta-neutral short-term straddles (Jan 2000-Dec 2002)

The sample consists of at-the-money (ATM) call (put) options defined as $X/S \geq 1.03$ ($X/S \leq 0.97$) and out-of-the-money (OTM) options defined as $0.97 < X/S < 1.03$ with maturity ranging from 6 – 60 days. Daily straddle trading is based on one day-ahead volatility forecasts from competing models. The average maturity of the options used in the analysis was 20.40 days and the average moneyness was 1.005. Daily returns have been adjusted for transaction costs and bid-ask bounce. We present the following metrics for profits from option trading: maximum, minimum, median, the first four moments of returns, p -values (based on Johnson's non-normal t -stats) for mean and median returns, ratio of positive returns vs. negative return days, the Sharpe ratio and the p -value robust to non-iid returns based on Lo, 2002. Details of the straddle trading are described in Appendix 3.

<i>Panel A: Model prices</i>												
	Market prices	IV	GARCH - L	RV	ARFIMA	ARFIMA- L	ARFIMA- LJ	IV+GARCH- L	IV+RV	IV+ ARFIMA	IV+ ARFIMA-L	IV+ ARFIMA-LJ
Calls	22.85	21.53	20.36	16.87	16.89	16.90	16.23	21.65	21.65	21.68	21.68	21.41
Puts	27.57	29.39	28.17	24.56	24.59	24.59	23.83	29.48	29.47	29.50	29.50	29.22
Straddles	50.42	50.92	48.53	41.43	41.48	41.49	40.05	51.13	51.12	51.18	51.19	50.63
<i>Panel B: Straddle trades</i>												
Total straddle trades		429	456	466	470	471	468	432	429	423	423	423
Long straddle trades		244	190	17	17	17	40	255	251	250	250	228
Short straddle trades		185	266	449	453	454	428	177	178	173	173	195
<i>Panel C: Percentage daily returns from straddle trades</i>												
Maximum		97.31	87.94	88.07	88.07	88.07	88.07	97.31	98.08	97.31	97.31	97.31
Minimum		-57.78	-57.68	-57.68	-57.68	-57.68	-57.68	-55.41	-55.41	-55.41	-55.41	-55.41
Mean		2.74	0.55	0.22	0.06	0.08	0.00	2.67	3.00	2.94	3.05	3.23
p -value of mean returns		0.01	0.58	0.82	0.95	0.94	1.00	0.01	0.00	0.00	0.00	0.00
Standard deviation		21.08	21.29	20.41	20.51	20.49	20.27	21.41	21.12	20.49	20.54	21.45
Skewness		1.27	0.49	-0.52	-0.53	-0.53	-0.66	1.16	1.41	1.36	1.34	1.14
Kurtosis		8.03	6.46	5.40	5.37	5.38	5.31	7.59	8.15	8.11	8.02	7.49
Positive vs. negative returns		1.21	1.05	1.51	1.49	1.50	1.60	1.20	1.17	1.19	1.21	1.29
Sharpe ratio		0.13	0.03	0.01	0.00	0.00	0.00	0.12	0.14	0.14	0.15	0.15
p -value of Sharpe ratio		0.01	0.61	0.82	0.95	0.94	0.99	0.02	0.01	0.01	0.00	0.00

Table 6

Sell and hedge strategy based on daily volatility forecasts (S & P 500 Index call options: Jan 2000- Dec 2002)

The sample consists of at-the-money (ATM) options defined as $X/S \geq 1.03$ and out-of-the-money (OTM) options defined as $0.97 < X/S < 1.03$ with maturity ranging from 6 – 60 days. All options within a sub-sample are sold based on specific model prices and delta hedged daily until maturity. We use one day-ahead volatility forecasts from competing models to arrive at model prices. We delta hedge based on one day-ahead delta forecasts from competing volatility models. Daily returns have been adjusted for transaction costs. We present the following metrics for profits from option trading: maximum, minimum, median, the first four moments of returns, p -values (based on Johnson's non-normal t -stats) for mean and median returns, ratio of positive returns vs. negative return days, the Sharpe ratio and the p -value robust to non-iid returns based on Lo, 2002.

	IV	GARCH-L	RV	ARFIMA	ARFIMA-L	ARFIMA-LJ	IV+GARCH-L	IV+RV	IV+ARFIMA	IV+ARFIMA-L	IV+ARFIMA-LJ
<i>Panel A: OTM options (N=678)</i>											
Maximum	1.012	0.968	0.766	0.835	0.842	3.511	0.922	0.908	0.832	0.848	1.443
Minimum	-0.964	-1.061	-1.282	-1.427	-1.406	-2.016	-1.010	-0.802	-0.969	-0.939	-1.062
Mean	-0.023	-0.023	-0.102	-0.090	-0.09	-0.042	-0.026	-0.029	-0.030	-0.030	-0.008
p -value of mean returns	0.067	0.046	0.001	0.001	0.001	0.016	0.029	0.017	0.011	0.011	0.548
Standard deviation	0.316	0.298	0.292	0.297	0.294	0.428	0.303	0.301	0.297	0.299	0.335
Skewness	0.294	-0.001	-0.522	-0.719	-0.708	0.578	0.180	0.263	0.098	0.141	0.407
Kurtosis	4.037	4.207	4.673	5.370	5.401	12.996	3.809	3.692	3.581	3.622	4.204
Sharpe ratio	-0.108	-0.116	-0.385	-0.342	-0.344	-0.124	-0.124	-0.133	-0.138	-0.138	-0.057
p -value of Sharpe ratio	0.842	0.842	0.544	0.610	0.612	0.861	0.821	0.808	0.796	0.800	0.912
<i>Panel B: ATM options (N=713)</i>											
Maximum	0.868	0.943	0.705	0.783	0.775	0.751	0.913	0.919	0.922	0.936	0.933
Minimum	-0.999	-1.023	-0.998	-1.024	-1.017	-1.047	-0.957	-0.987	-1.000	-0.990	-1.002
Mean	-0.027	-0.047	-0.105	-0.091	-0.091	-0.086	-0.026	-0.028	-0.027	-0.026	-0.026
p -value of mean returns	0.026	0.001	0.001	0.001	0.001	0.001	0.028	0.023	0.027	0.033	0.032
Standard deviation	0.314	0.305	0.289	0.291	0.290	0.290	0.314	0.316	0.315	0.315	0.317
Skewness	-0.128	-0.081	-0.313	-0.238	-0.234	-0.265	-0.106	-0.054	-0.089	-0.051	-0.06
Kurtosis	3.537	3.651	3.692	3.715	3.701	3.749	3.392	3.539	3.512	3.488	3.463
Sharpe ratio	-0.121	-0.191	-0.400	-0.351	-0.353	-0.336	-0.12	-0.122	-0.120	-0.117	-0.117
p -value of Sharpe ratio	0.810	0.719	0.479	0.533	0.532	0.551	0.807	0.806	0.810	0.814	0.811

Figure 1

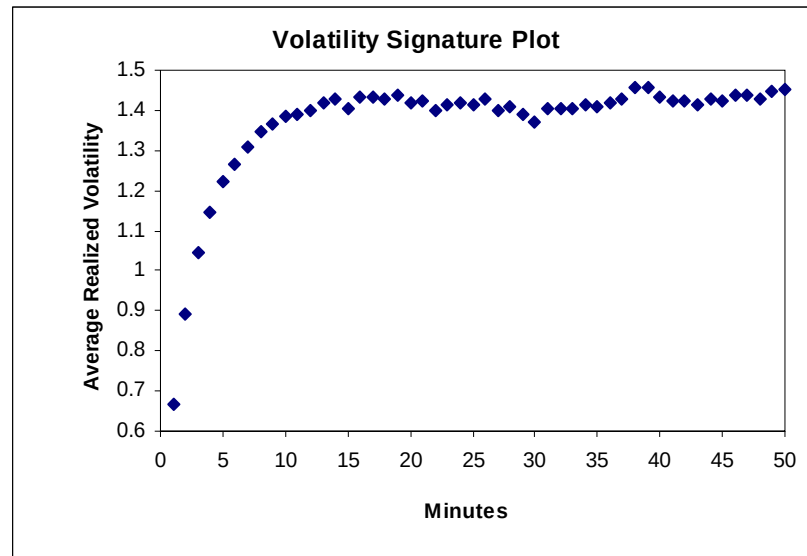


Figure 2

